Curriculum Vitae

Supriyo Sarkar



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PROFILE SUMMARY

I am an M.Tech graduate in Computer Science and Engineering with research interests and technical expertise in Machine Learning, Deep Learning, Federated Learning, Natural Language Processing (NLP), and LLMs. My academic background and research experience have strengthened my ability to identify research gaps, investigate challenging problems, develop effective methodologies, and make meaningful contributions to the field. I am passionate about exploring emerging research directions, advancing my technical expertise, and developing innovative solutions to address real-world challenges through impactful research.

RESEARCH INTERESTS

Machine Learning, Deep Learning, Federated Learning, Natural Language Processing (NLP), LLMs, Medical Image Analysis, Knowledge Distillation and Model Compression, Homomorphic Encryption, Artificial Intelligence, Quantum Computing.

EDUCATIONAL QUALIFICATIONS

Master of Technology in Computer Science and Engineering Kalyani Government Engineering College **[MAKAUT, WB]** CGPA: 8.15/10	[2024 - 2026] [Nadia, WB]
Bachelor of Technology in Computer Science and Engineering Regent Education & Research Foundation **[MAKAUT, WB]** CGPA: 8.95/10	[2020 - 2024] [Kolkata, WB]
12th - West Bengal Council of Higher Secondary Education (W.B.C.H.S.E) Mal Adarsha Bidya Bhaban Marks: 70.10%	[2019] [Malbazar, Jalpaiguri, WB]
10th - West Bengal Board of Secondary Education (W.B.B.S.E) Banarhat High School Marks: 68.71%	[2017] [Banarhat, Jalpaiguri, WB]

TECHNICAL KNOWLEDGE & SKILLS

- **Programming Languages:** Python, C, C++
- **Data Science:** Machine Learning, Deep Learning, Natural Language Processing (NLP), LLMs, Federated Learning, Medical Image Analysis, Model Compression and Knowledge Distillation
- **Tools:** Jupyter Notebook, VS Code, Git, GitHub, MS Office
- **Core Computer Science:** Data Structures and Algorithms, Operating Systems, Computer Networks, Database Management Systems
- **Extra Knowledge:** Quantum Computing, Arduino IDE
- **Soft Skills:** Communication, Teamwork, Problem-solving, Time management

M.TECH FINAL YEAR DISSERTATION

- **Privacy-Preserving Alzheimer's Disease Detection for Non-IID MRI Data: A Comparative Analysis of Federated Learning Algorithms under Homomorphic Encryption.**

  Skills: Python, Federated Learning, Non-IID Data Distribution, Homomorphic Encryption, Deep Learning and Medical Image Analysis

  Description: Research on Alzheimer's disease classification from MRI scans using Federated Learning. Compared MobileNetV2, EfficientNetB0 and a proposed Custom CNN with FedAvg, FedProx and FedDyn across nine configurations, using four-class MRI data and five clients with strongly Non-IID Dirichlet partitions over 30 communication rounds. The Custom CNN with Group Normalisation and FedAvg achieved 94.59% test accuracy, 94.80% macro F1-score and 99.50% macro AUC in simulation. CKKS homomorphic encryption was prototyped with TenSEAL on sampled parameters, and full encrypted aggregation is part of ongoing follow-up work.

B.TECH FINAL YEAR DISSERTATION

- **Quantum Gate Performance Assessment in the Presence of Superposition State using IBMQ Server**

  Skills: Python, Quantum Computing, Git and GitHub, MS Office

  Description: Measuring the efficiency of the quantum gate with the superposition state of IBMQ Server to enhance the technique of mitigations of noise, reducing the processor heat of the NISQ devices of IBM Quantum Computing Servers for more practical quantum computation, therefore increasing overall effectiveness and dependability of the device.

SELECTED PROJECTS

- **DeveloperBro. - Portfolio Website**
  Skills: HTML/CSS, JavaScript, Git and GitHub
  Demonstrating "**DeveloperBro.**" a dynamic portfolio website which is blend of professionalism and creativity.

- **Arduino Uno Third Hand Eye Project for the visually impaired!**
  Skills: C/C++, Arduino Uno, Arduino IDE
  This project focuses on recognizing obstacles in the surroundings and alerting the user through audio feedback, boosting their mobility and safety.

COURSES & CERTIFICATIONS

- Learn C++ Programming - Beginner to Advanced: Deep Dive in C++ (Udemy)
- Crash Course on Python (Google/Coursera)
- The Joy of Computing Using Python (NPTEL)
- Python for Data Science (NPTEL)
- Deep Learning (NPTEL)
- Soft Skills (NPTEL)
- CSS, Bootstrap, JavaScript and PHP Stack Complete Course (Udemy)

ACHIEVEMENTS

- **Smart India Hackathon 2022 -** Selected for develop "**Tour India**" travel website.
- **Qualified WB PGET 2025** with **Rank 43**.

STRENGTH & HOBBIES

- Hard & Smart Working
- Honest & Punctual
- Good Communication skill
- Photography, Cricket, Sci-Fi Movies, Cooking, Travelling

LANGUAGES

- English - Moderate proficiency
- Bengali - Professional proficiency
- Hindi - Professional proficiency